About this Monthly Newsletter

Welcome to the first IE Lung patient newsletter! In an effort to share our collective >100 years of pulmonary and critical care experience among 7 physicians, each month, we will shed light on common and uncommon problems regarding breathing, sleep, and your health through our website and patient portal.

This way you will get introduced to all seven excellent pulmonologists and their areas of expertise.

This month, we have Dr. Giri who would like to focus on a common question:

“Why do I feel short of breath?”

Understanding Shortness of Breath

This month with Dr. Paresh Giri, MD, FCCP



Breathing is life. Often, we don't even notice we are breathing. But when breathing becomes difficult, we are certainly more aware of it. The science of breathing is fascinating and the reasons we feel out of breath is even more intricate and complex.

1. Sometimes I have difficulty breathing. What could be causing it?

Feeling short of breath means breathing feels hard or you cannot seem to get enough air. I often start by looking at four possible causes. More than one may affect you at the same time.

1. **The heart.** Your heart pumps blood that carries oxygen through your body. If it is not pumping well, or has a valve or heartbeat problem, you may feel breathless. We may suggest an ultrasound of your heart or a visit with a heart specialist.
2. **The lungs.** Asthma, long-term lung disease such as COPD, lung scarring, and problems with blood vessels in the lungs can make breathing harder. Breathing tests help us check how well your lungs work.
3. **Loss of strength and fitness.** Being less active can weaken your muscles. Walking or climbing stairs may then take more effort. We call this deconditioning. After checking the cause, we can help you plan a safe, gradual return to activity.
4. **Anemia.** Your red blood cells carry oxygen using a protein called hemoglobin. If you do not have enough, you may feel tired or out of breath. A blood test can check for anemia. Treatment depends on what is causing it.

When to get urgent help

Call 911 for sudden or severe trouble breathing, especially with chest pain, fainting, confusion, blue or gray lips, or trouble speaking in full sentences. Do not wait for a reply through the patient portal.

For new or worsening breathlessness, contact your healthcare team promptly

2 How do you test my lungs?

At IE Lung, we offer breathing tests called pulmonary function tests, or PFTs. You sit and breathe through a mouthpiece. A staff member guides you through each step. You may be asked to take a deep breath and blow out hard. Some tests use a clear booth.

These tests check how much air your lungs hold and how quickly you can blow air out. Another part helps us check how well oxygen can move from your lungs into your blood.





Examples of a seated breathing test and an exercise test. Actual equipment may differ.

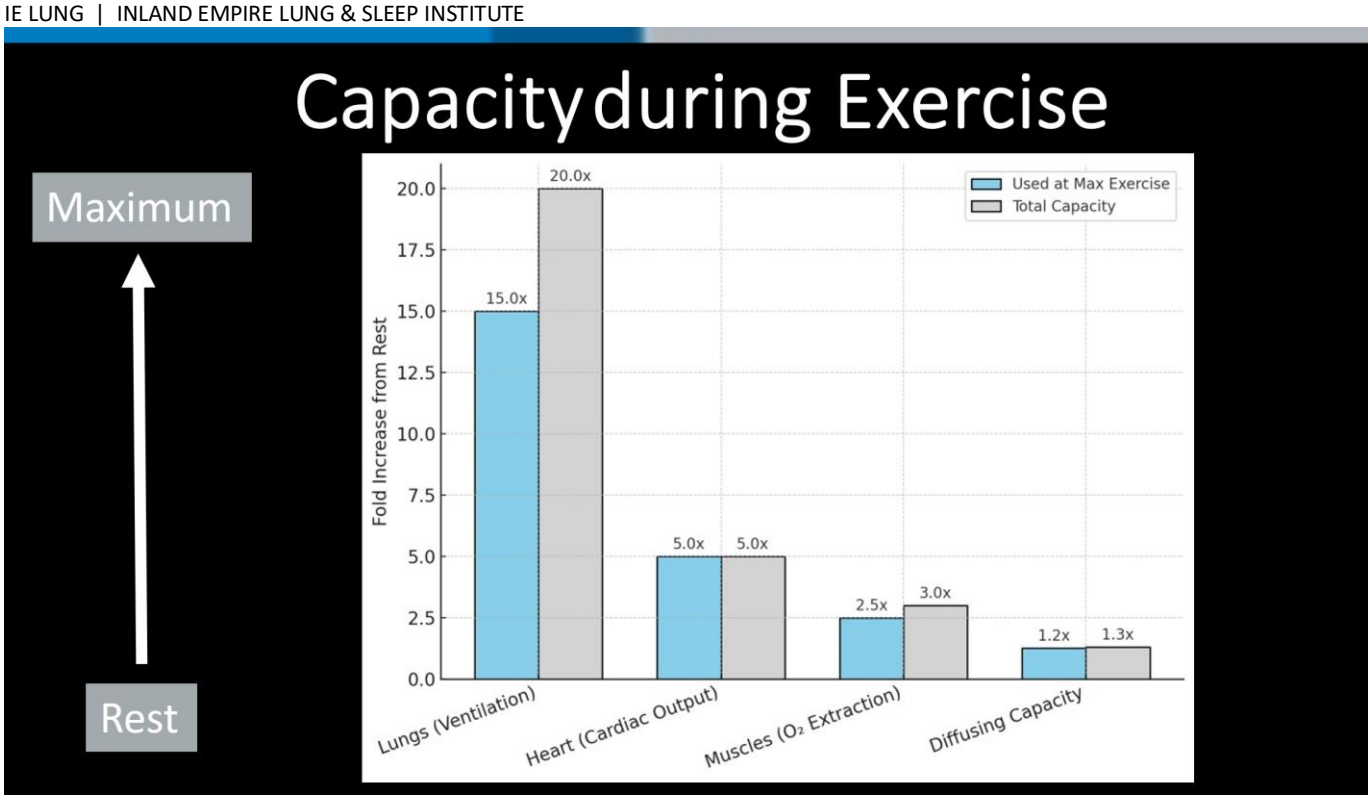
3 What if I get breathless when I walk or exercise?

A test done while you sit may not tell the whole story. We also offer an exercise breathing test called cardiopulmonary exercise testing, or CPET. It shows how your heart, lungs, and muscles work together when you are active.

You pedal an exercise bike while wearing a mask that measures the air you breathe. Staff check your heartbeat, blood pressure, and oxygen level and your VO_2 max. The results help us understand what makes activity hard for you.

During activity, your muscles need more oxygen, so you breathe faster and your heart pumps more blood. Healthy lungs often have extra breathing capacity.

The lungs are amazing organs! They have the capacity to increase by more than 15 times their capacity at rest to help you breathe while doing heavy exertion, see figure below. Other organs like the heart and muscles have much lesser exercise capacity. (That is why we are lung doctors!! Ahem! The most important organ in the body!)



4 How can I prepare for my visit?

Write down when you feel breathless and what you are doing when it happens. Tell us about any cough, wheezing, chest discomfort, or swelling. Bring your medicine list and tell us what your goals are and what you would like to do more easily, such as walking or climbing stairs.

Together, we will choose the next steps together based on your symptoms and exam.

5 Can breathing exercises and physical exercises help me breathe better?

Absolutely. Though there may be times to be cautious if you have significant lung or heart problems. We usually recommend monitored pulmonary rehabilitation program for such conditions. But for many people daily exercise of around 30 minutes with a combination of cardio, resistance and weight is ideal to reach your optimum pulmonary health.

This newsletter provides general information. Talk with your healthcare team about your own symptoms and care